Amirmohammad (Amir) Mohammadi

Computer Engineering PhD candidate · Texas A&M University

College Station, Texas · U.S. Permanent Resident

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Research focus: Data- and parameter-efficient learning for real-world sensing.

Education

PhD in Computer Engineering · Texas A&M University

September 2022 – present · Expected December 2026

Thesis: Distribution-guided Parameter-efficient Fine-tuning of Foundation Models for Waveform-based Sensing

MS in Electrical Engineering · Sharif University of Technology

September 2018 – February 2021

Thesis: Design and Evaluation of an Integrated Biomedical System for Human Stress Detection and Monitoring

BS in Electrical Engineering · University of Tabriz

September 2014 – September 2018

Research experience

Graduate Research Assistant · Texas A&M University

September 2022 – present

- Developed distribution-aware, parameter-efficient adaptation methods for Transformer foundation models in waveform-based sensing.
- Built a few-shot whole-slide image classification pipeline to study parameter-efficient adaptation of vision-language models for liver histology.
- Designed time-frequency feature combinations for histogram-layer neural networks in acoustic classification.
- Established a cross-domain transfer benchmark in which ImageNet-pretrained models outperformed audio-pretrained models by 3 percentage points on the evaluated acoustic classification task.
- Developed physics-informed models for cuffless blood-pressure estimation; the collaborative study reduced ground-truth training data requirements by an average factor of 15 on its evaluated datasets.

Graduate Student Researcher · Sharif University of Technology

July 2019 – February 2021

- Designed an integrated ECG/EDA wearable sensor, from circuitry and Bluetooth Low Energy electronics to machine-learning algorithms.
- Achieved 94% stress-detection accuracy in an 18-participant study and threefold battery life relative to the evaluated alternatives.

Technical skills

Machine learning: PyTorch, Hugging Face Transformers, scikit-learn, foundation-model adaptation, physics-informed learning, multimodal learning

Scientific computing: Python, NumPy, pandas, Matplotlib, MATLAB, C

Research engineering: Git, Linux, Jupyter, high-performance computing, waveform and time-frequency analysis, wearable sensing

Research support: Doctoral research assistantship support from MIT Lincoln Laboratory, the National Institutes of Health, and the Texas A&M Department of Electrical & Computer Engineering.

Publications & research contributions

Peer-reviewed & accepted publications

J. Ritu, **A. Mohammadi**, D. Carreiro, A. Van Dine, J. Peeples. **Structural and Statistical Audio Texture Knowledge Distillation for Acoustic Classification**. *IEEE Access (2026)*. Preprint · PDF

A. Mohammadi, D. Carreiro, A. Van Dine, J. Peeples. **Histogram-based Parameter-efficient Tuning for Passive and Active Sonar Classification**. *IEEE International Geoscience and Remote Sensing Symposium (IGARSS) (2026) · Accepted*. Preprint · PDF · Code

F. Orvati Nia, **A. Mohammadi**, S. Al Kharsa, P. Naikare, Z. Hampel-Arias, J. Peeples. **Neighborhood Feature Pooling for Remote Sensing Image Classification**. *IEEE/CVF WACV Workshops (2026), pp. 1393–1402*. Paper

A. Mohammadi, I. Masabarakiza, E. Barnes, D. Carreiro, A. Van Dine, J. Peeples. **Investigation of Time-Frequency Feature Combinations with Histogram Layer Time Delay Neural Networks**. *OCEANS 2025 Brest (2025), pp. 1–6*. Publisher

A. Mohammadi, T. Kelhe, D. Carreiro, A. Van Dine, J. Peeples. **Cross-Domain Knowledge Transfer for Underwater Acoustic Classification Using Pre-trained Models**. *OCEANS 2025 Brest (2025), pp. 1–6*. Publisher

K. Sel, **A. Mohammadi**, R. I. Pettigrew, R. Jafari. **Physics-informed neural networks for modeling physiological time series for cuffless blood pressure estimation**. *npj Digital Medicine (2023), vol. 6, article 110*. Paper · PDF

A. Mohammadi, M. Fakharzadeh, B. Baraeinejad. **An Integrated Human Stress Detection Sensor Using Supervised Algorithms**. *IEEE Sensors Journal (2022), vol. 22, no. 8, pp. 8216–8223*. Publisher

Manuscripts under review

A. Mohammadi, J. Peeples, A. Van Dine. **Parameter-efficient Dual-encoder Architecture with Differentiable Choquet Integral Fusion for Underwater Acoustic Classification**. *arXiv:2606.02341 (2026) · Under review*. Preprint · PDF

A. Mohammadi, R. Khadka, K. A. Grant, M. L. Benton, J. Peeples. **Predicting Ethanol Drinking Status from Nonhuman Primate Liver Histology: A Proof of Concept for Parameter-efficient Fine-tuning in Few-Shot Vision-Language Modeling**. *Manuscript (2026) · Under review*.

Academic service, teaching & mentoring

Reviewer, IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). Reviewer, Expert Systems with Applications. Session chair, IEEE International Geoscience and Remote Sensing Symposium (IGARSS).

Mentored undergraduate and graduate students on end-to-end AI projects. Delivered guest lectures in Machine Learning for Electrical Engineering at Texas A&M University. Showcased AI projects at Night at the Zach during the Texas Science & Engineering Fair.

Awards

IEEE Geoscience and Remote Sensing Society travel grant · May 2026

Texas A&M Electrical & Computer Engineering conference travel grant · May 2026

Languages: English (proficient), Farsi (native/bilingual), Azerbaijani Turkish (native/bilingual)